

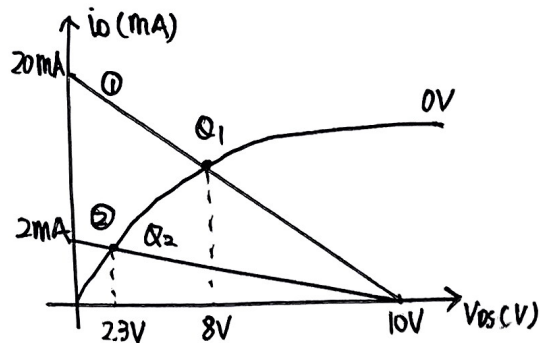
1. 场效应管

(1). 绝缘栅型, N 沟道型耗尽型场效应管.

(2). $V_{GS} = V_{DD} - I_D \cdot R$.

流过电阻 R 的电流为 I_D .

(3). $V_{GS} = 0V$. $V_T = -3V$.



① 当 $R = 500\Omega = 0.5k\Omega$ 时, $i_D = \frac{V_{DD}}{R} - \frac{1}{R} V_{DS}$
 $= 20 - 2V_{DS}$

② 当 $R = 5k\Omega$ 时, $i_D = 2 - 0.2V_{DS}$.

Q_1 在放大区, Q_2 在可变电阻区工作.

$V_{GS} - V_{GS2} = -2.3 > -3 = V_T$

2. 三极管的静态和动态分析

(1). 静态分析:

对 T_2 : $V_{GS} = 0$.

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_T}\right)^2 = I_{DSS}$$

$$V_{DS} = V_E$$

对 T_1 : $I_E = I_D = I_{DSS}$.

$$I_B = \frac{I_E}{\beta + 1} = \frac{1}{1 + \beta} I_{DSS}$$

$$I_C = \beta I_B = \frac{\beta}{1 + \beta} I_{DSS}$$

$V_{BE} = 0.7V$, 左侧戴维宁等效.

$$V_B = V_{CC}' - I_B R_2' = \frac{R_2}{R_1 + R_2} V_{CC} - \frac{R_1 \parallel R_2}{1 + \beta} I_{DSS}$$

$$V_E = V_B - V_{BE} = \frac{R_2}{R_1 + R_2} V_{CC} - \frac{R_1 \parallel R_2}{1 + \beta} I_{DSS} - V_{BE}$$

$$V_{DS} = V_E = \frac{R_2}{R_1 + R_2} V_{CC} - V_{BE}$$

$$V_{CE} = V_{CC} - I_C R_C - V_E$$

$$= V_{CC} - \frac{\beta}{1 + \beta} I_{DSS} R_C - \frac{R_2}{R_1 + R_2} V_{CC} + \frac{R_1 \parallel R_2}{1 + \beta} I_{DSS} + V_{BE}$$

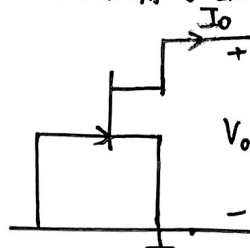
$$= \frac{R_1}{R_1 + R_2} V_{CC} - \frac{\beta}{1 + \beta} I_{DSS} R_C + \frac{R_1 \parallel R_2}{1 + \beta} I_{DSS} + V_{BE}$$

(2). CB 电路,

I_2 是电流源电路.

(3). 动态分析:

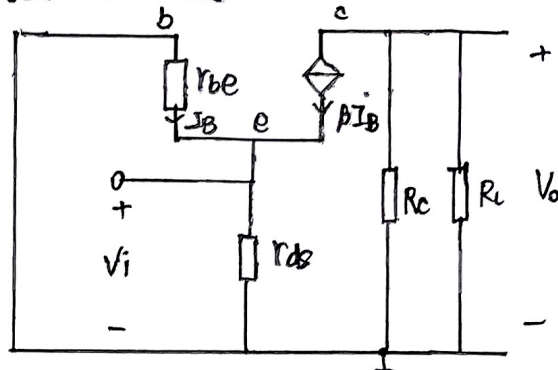
先分析电流源电路:



$$I_O = I_{DSS}$$

$$V_O = V_{DS}$$

则上面的微变等效电路图:



$$A_v = \frac{-\beta I_B (R_C \parallel R_L)}{-I_B r_{be}} = \frac{\beta (R_C \parallel R_L)}{r_{be}}$$

$$R_i = r_{ds} \parallel \frac{I_B r_{be}}{(\beta + 1) I_B} = r_{ds} \parallel \frac{r_{be}}{1 + \beta}$$

$$R_o = R_C$$

3. 分立元件放大电路分析

第I级是CC, 第II级是CB.

(1). 静态分析.

$$V_{EQ} = V_{BQ} - V_{BE} = -0.7V$$

由 T_1, T_2 特性一致, 知: $V_{BE1Q} = V_{BE2Q} = 0.7V$.

$$I_{B1Q} = I_{B2Q}$$

$$I_{C1Q} = I_{C2Q}$$

$$I_{E1Q} = I_{E2Q}$$

$$I_{E1Q} + I_{E2Q} = \frac{V_{EQ} + V_{EE}}{R_E} = 1mA.$$

$$I_{E1Q} = I_{E2Q} = 0.5mA.$$

$$I_{C1Q} = I_{C2Q} = \frac{\beta I_{E1Q}}{1+\beta} = \frac{\beta \cdot 0.5}{1+\beta} mA \approx 0.5mA.$$

$$V_{CE1Q} = V_{CC} - V_{EQ} = 6.7V$$

$$V_{CE2Q} = V_{CC} - I_{C2Q} R_C - V_{EQ} = 6 - 0.5 \times 2 + 0.7 = 5.7V$$

(2). 动态分析. (由于是差分电路, 故不作近似处理)

第I级: CC电路.

$$A_{vI} = \frac{(1+\beta) R_E \parallel R_{iI}}{r_{be} + (1+\beta) R_E \parallel R_{iI}} \quad R_{iI} = R_{iII} = \frac{r_{be2}}{1+\beta}$$

$$R_i = r_{be1} \parallel (1+\beta) R_E \parallel R_{iI}$$

第II级: CB电路.

$$A_{vII} = \frac{\beta R_{iI} \parallel R_C}{r_{be2}} = \frac{\beta R_C}{r_{be}}$$

$$R_{oII} = R_C$$

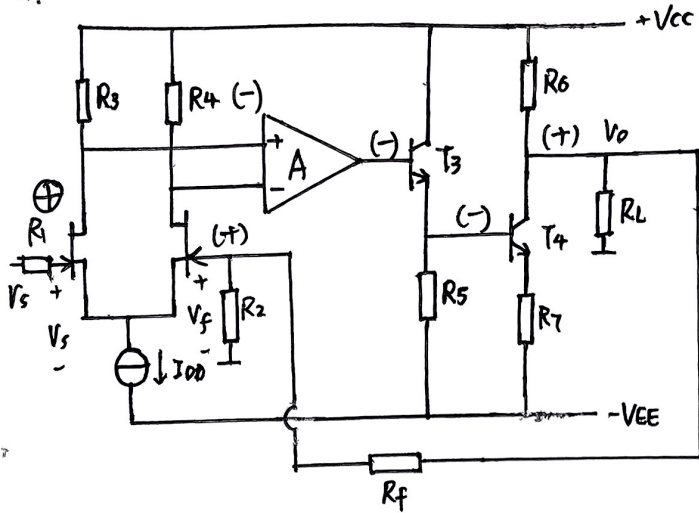
$$\text{综上, 最终 } A_v = A_{vI} \cdot A_{vII} = \frac{(1+\beta) R_E \parallel \frac{r_{be}}{1+\beta}}{r_{be} + (1+\beta) R_E \parallel \frac{r_{be}}{1+\beta}} \cdot \frac{\beta R_C}{r_{be}}$$

$$= \frac{(1+\beta) R_E}{2(1+\beta) R_E + r_{be}} \cdot \frac{\beta R_C}{r_{be}}$$

$$R_o = R_{oII} = R_C$$

反馈类型判断

(1).



串联反馈, $V_f = \frac{R_2}{R_2 + R_f} V_o$. 电压反馈

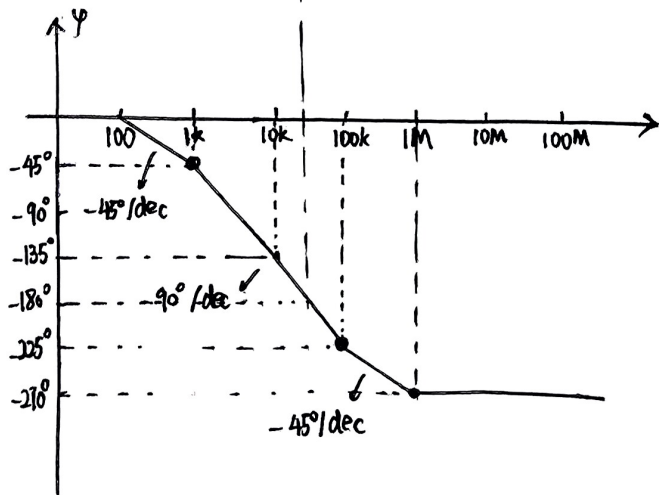
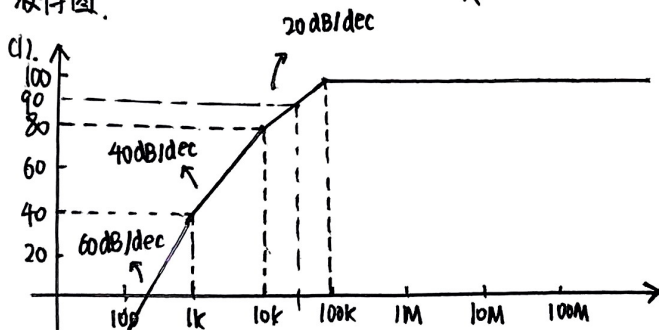
反馈

(2). 将电阻 R_f 接入到差分放大电路的1端口, 同时将最后一个CE电路改成CB电路构成负反馈.

$$(3). A_{vf} = \frac{V_o}{V_f} = 1 + \frac{R_f}{R_2}$$

串联反馈, 故 $R_{if} = \infty$.

波特图.



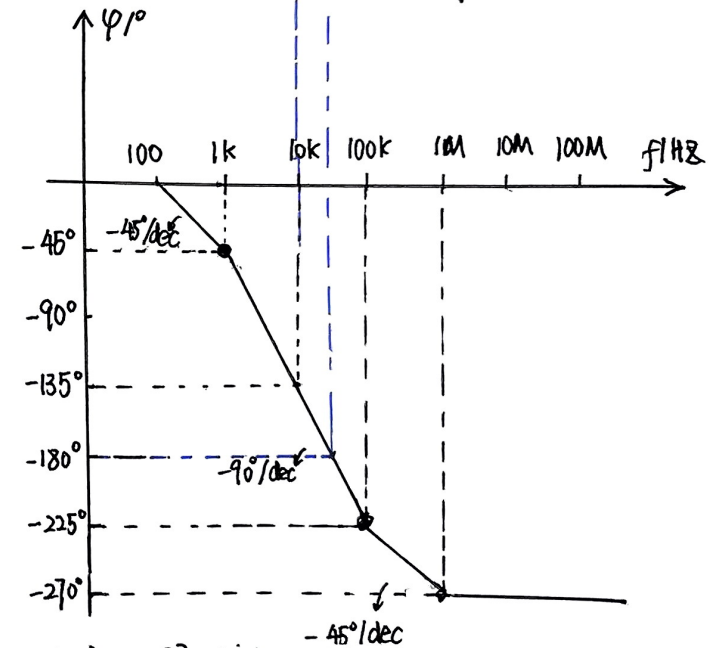
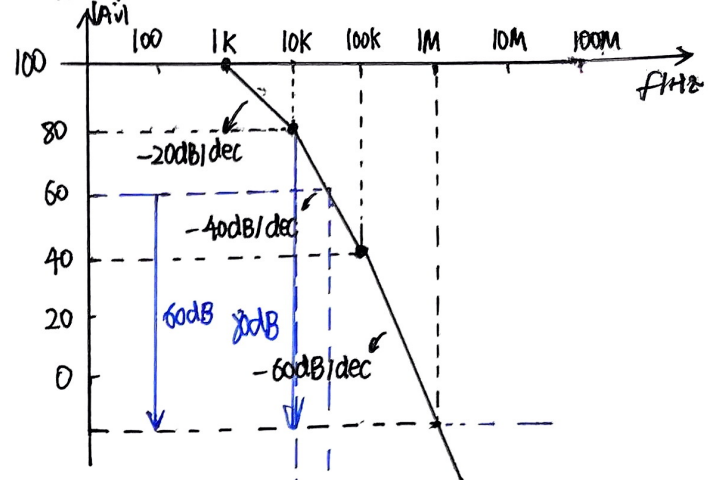
$$(2). |F| = 10^{-4.5}$$

$$|A_{vf}| = \frac{1}{|F|} = 10^{4.5}$$

$$(3). |F|_{\max} = 10^{-4}$$

最大可取 10^{-4}

(1). 波特图:



(2). 当 $F = 10^{-3}$ 时, 产生临界振荡

$$|A| = 10^3, |F| = 10^{-3}, A_{vf} = \frac{A}{1 + AF} = \frac{10^3}{2} = 500$$

(3). 当 $F = 10^{-4}$ 时, 产生 45° 相位裕度

$$\text{则 } F_{\max} = 10^{-4}$$

功率电路

(1) 互补对称共集电路。\$R_4, R_5\$ 用于给三极管提供合适的静态偏置，\$D_1, D_2\$ 用于防止交越失真并提供偏置电压。

$$(2) V_{om1} = V_{CC} - V_{CES} = 15 - 1 = 14V$$

$$V_{om2} = V_{O_{Pmax}} + V_p - V_{BE} = 13 + 0.7 - 0.7 = 13V$$

$$V_{om} = V_{om2} = 13V$$

$$P_{om} = \frac{V_{om}^2}{2R_L} = \frac{13^2}{2 \times 10} = 8.45W$$

$$P_{em} = \frac{2V_{CC}V_{om}}{\pi R_L}$$

$$\eta = \frac{P_{om}}{P_{em}} = \frac{\pi V_{om}}{4 V_{CC}} = \frac{\pi}{4} \times \frac{13}{15} = 68.069\%$$

(3) 提高输入电阻 \$\rightarrow\$ 串联反馈

稳定输出电压 \$\rightarrow\$ 电压反馈

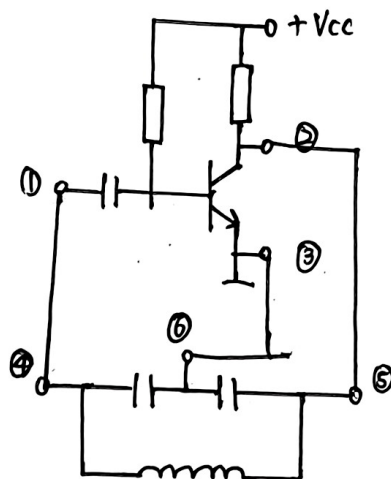
$$(4) A_{vf} = \frac{4}{0.1} = 40$$

$$A_{vf} = \frac{V_o}{V_i} = \frac{V_o}{V_f} = \frac{R_f + R_2}{R_2}$$

$$\therefore R_f = R_2 (A_{vf} - 1) = 39k\Omega$$

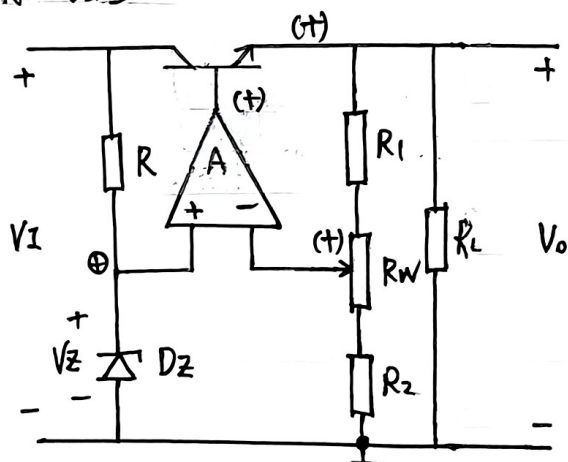
$$I_{OLmax} = I_{om} - \frac{V_{om1}}{R_1 + R_3 + R_w}$$

$$= 101 - \frac{8.4}{1+2+3} = 99.6mA$$



①④, ②⑥, ③⑤

稳压电路



$$(1) V_f = \frac{R_1 + \frac{1}{2}R_w}{R_2 + \frac{1}{2}R_w} V_o \Rightarrow V_o = 12V$$

$$(2) V_+ = +V_Z, V_- = +V_Z$$

$$(3) \text{当 } W \text{ 在上端时, } V_{om1} = \frac{R_w + R_1 + R_2}{R_w + R_2} V_Z = (1 + \frac{R_1}{R_w + R_2}) V_Z$$

$$= (1 + \frac{1}{5}) \times 7 = +8.4V$$

$$\text{当 } W \text{ 在下端时, } V_{om1} = \frac{R_w + R_1 + R_2}{R_2} V_Z = (1 + \frac{R_1 + R_w}{R_2}) V_Z$$

$$= (1 + \frac{1+3}{2}) V_Z = +21V$$

$$V_{om2} = V_{Amax} - V_{BE} = 19.3V$$

$$I_{om1} = (A_{IP}) I_{Amax} = 101 \times 1 = 101mA$$

$$\therefore V_{omax} = 19.3V$$

$$(4) I_{om} = 101mA$$

三角波发生器:

(1). A_2 处于非线性工作状态.

(2). 同相输入的滞回比较器.

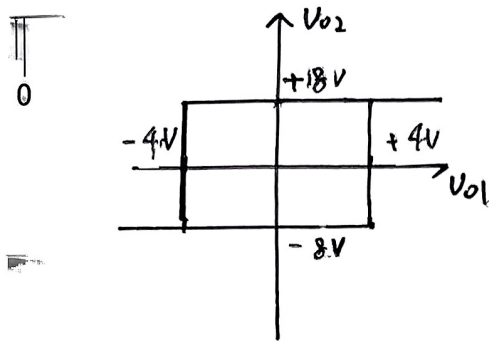
两个阈值电压: $\frac{V_{01}-V_+}{R_2} + \frac{V_{02}-V_+}{R_3} = 0$

$$\Rightarrow V_+ = \frac{R_3}{R_2+R_3} V_{01} + \frac{R_2}{R_2+R_3} V_{02} = V_- = 0V.$$

$$V_{TH} = -\frac{R_2}{R_3} V_{02} = -\frac{R_2}{R_3} \times -8 = +4V$$

$$V_{TL} = -4V$$

当 $V_{01} > V_{TH}$ 时, $V_+ > V_-$, 输出高电平 $V_{02} = +8V$



(3). ① 设 $t=0$ 时刻, $V_{02} = +8V$, $V_{01} = -V_C(0+) = 0V$.

对 A_1 分析, 由虚短虚断知: 电容在充电

$$V_C(t) = \frac{V_{02}}{RC} t, \quad V_{01} = -V_C(t) = -\frac{V_{02}}{RC} t.$$

则 V_{01} 减小, V_{02} 保持 $+V_2$ 不变. V_+ 减小

$$V_+ = \frac{R_3}{R_2+R_3} V_{01} + \frac{R_2}{R_2+R_3} V_{02} \Rightarrow 0.$$

V_+ 减小至 0 时, 电压开始翻转 $V_{01} = -\frac{R_2}{R_3} V_{02} = -4V$.

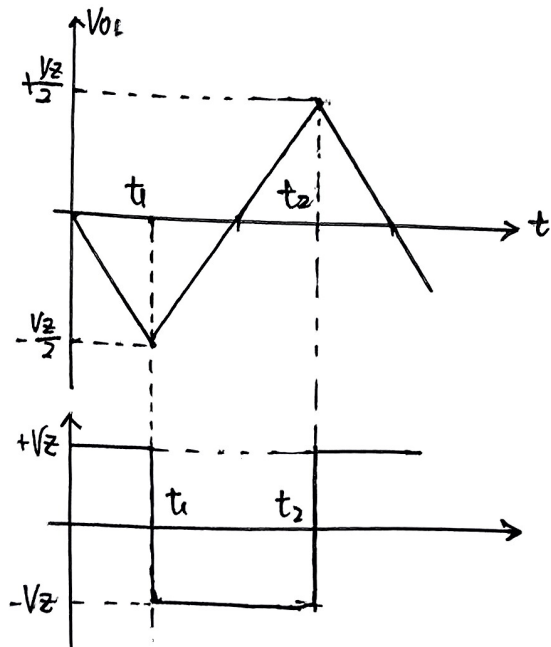
② $t=t_1$ 时刻, $V_2 = -8V$, $V_{01} = -4V$.

对 A_1 分析, 电容放电: $V_C(t) = \frac{V_2}{2} + \frac{V_{02}}{RC} t$.

$$V_{01}(t) = -\frac{V_2}{2} + \frac{V_2}{RC} t.$$

则 V_{01} 增大, V_{02} 保持 $-V_2$ 不变, V_+ 增大.

V_+ 增大至 0 时, 电压开始翻转. $V_{01} = +4V$.



(4). 若提高 V_{REF} , 则充电的速度减慢, 放电速度加快
对 V_{01} 而言, 下降直线变缓, 上升直线变陡.

对 V_{02} 而言, 高电平时间变长, 低电平时间变短

因此占空比提高